$$E_{\mathrm{V}} = \rho_{\mathrm{liqu}} V_{\mathrm{plasma}} [\mathrm{c_p}(T_2 - T_1) + \Delta H_{\mathrm{vap}}], \tag{10}$$

with c_{p} = 4.187 kJ K^{-1} kg^{-1}, T_2 = 100°C, T_1 = 20°C, and ΔH_{vap} = 2256 kJ kg^{-1}.

The bubble energy was obtained from Eq. (8), and the shock wave energy estimated as

$$E_{SW} = E_{abs} - E_V - E_B. \tag{11}$$

3.6. Determination of plasma energy density, electron density, pressure, and temperature

The *average plasma energy density* is given by the ratio of absorbed laser energy and plasma volume

$$U_{\mathrm{avg}} = \frac{E_{\mathrm{abs}}}{V_{\mathrm{plasma}}}. \tag{12}$$

The E_{abs} and V_{plasma} data had to be determined in separate measurement series because the 63x NA = 0.9 objective used for the transmission measurements could not be combined in confocal arrangement with the 20x NA = 0.5 objective for plasma photography. To determine the dependence of U_{avg} on pulse energy, functions were fitted through the V_{plasma} (E_{L}) and E_{abs}(E_{L}) data (Jandel Scientific, TableCurve2D), and U_{avg}(E_{L}) curves were calculated by dividing the fit functions.

For fs breakdown, the average *free electron density* ρ_c can be deduced from U_{avg} because here only one set of free electrons is produced, as recombination during the pulse plays no significant role. The average energy of CB electrons is given by the sum of effective ionization energy and kinetic energy $\varepsilon_{\mathrm{avg}} = \tilde{\Delta} + \varepsilon_{\mathrm{kin,avg}}$. It was shown to be constant for irradiance values between optical breakdown threshold and full ionization of the 1b$_1$ level [149] and can be approximated by $\varepsilon_{\mathrm{avg}} = (9/4)\tilde{\Delta}$ [22,37]. This yields

$$\rho_{\mathrm{c}} = \frac{U_{\mathrm{avg}}}{(9/4)\tilde{\Delta}}. \tag{13}$$

In order to assess laser plasma coupling, it is of interest to compare ρ_c with the critical electron density ρ_{crit} at which the plasma frequency $\omega_{\mathrm{p}} = \sqrt{\rho_{\mathrm{c}} e^2 / m_{\mathrm{e}} \varepsilon_0}$ equals the laser frequency ω_L [6,115,151]. Here e denotes the electron charge, m_{e} its mass, and ε_0 is the vacuum dielectric permittivity. Above ρ_{crit}, the electron movement can follow the electric field oscillations, and both reflectivity and absorptivity increase [151,236]. The *critical free-electron density* is given by

$$\rho_{\mathrm{crit}} = \omega_{\mathrm{L}}^2 \frac{m_{\mathrm{e}}\, \varepsilon_0}{e^2}. \tag{14}$$

The energy density data are also employed for calculating *plasma pressure and temperature* from EOS data, assuming isochoric energy deposition. The absorbing volume remains constant during energy

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deposition if neither thermal nor hydrodynamics expansion occur. Since thermal expansion proceeds with sound velocity, thermoelastic stress cannot relax, and isochoric energy deposition is stress-confined. The criterion for stress confinement is that an acoustic transient cannot completely traverse the absorbing volume during the laser pulse duration [22,188]. For 10-ns laser pulses and a sound speed of 1500 m/s, this the case for plasma diameters ≥ 15 µm. Since this criterion holds for all luminescent ns plasmas observed in this paper, we can use EOS data to estimate pressure and temperature of luminescent plasmas for the entire range of investigated pulse durations from femto- to nanoseconds.

Figure 3 illustrates the dependence of pressure and temperature on internal energy given by three different EOS. The IAPWS-95 formulation is a generally accepted reference for temperatures up to 1273 K and pressures up to 1000 MPa (10 kbar) [230]. It can be extrapolated also to higher internal energies but this extrapolation does not explicitly consider the dissociation of water molecules occurring at high temperatures. Dissociation is included in the SESAME table 7150 [237] and in the quantum molecular dynamics (QMD) calculations for high-density water by Mattsson and Desjarlais [172,173].

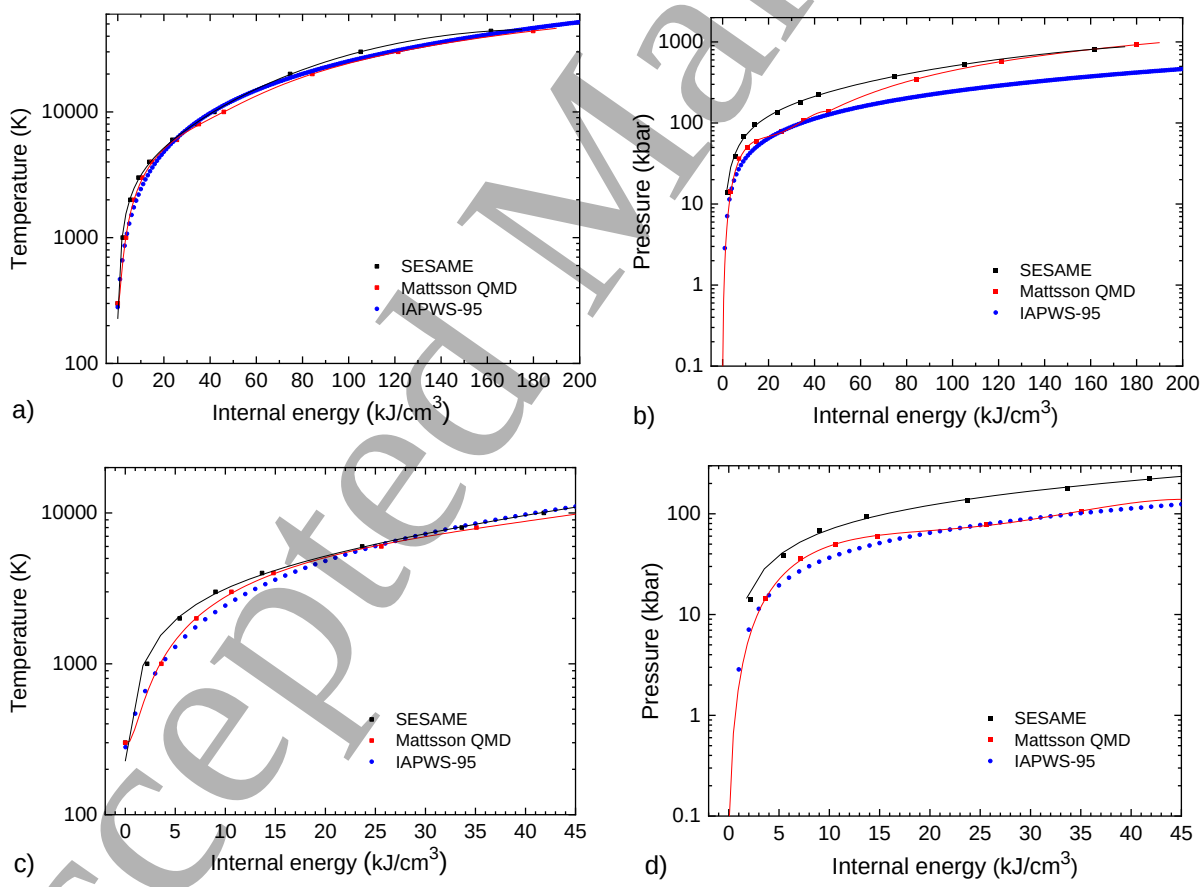

Figure 3 Water temperature and pressure as a function of internal energy U after isochoric energy deposition into liquid water (ρ_{water} = 998 kg/m^3) at room temperature. $T(U)$ and $p(U)$ plots are shown for the full available data range in (a) and (b), and as zoom for internal energies up to 45 kJ/cm^3 in (c) and (d), respectively. Each graph contains values based on the IAPWS-95 formulation of the EOS of water [230], data from SESAME table 7150

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[237], and QMD calculations from Refs. [172,173]. Differences between the EOS data are mainly related to the way in which water dissociation is considered (see text).

For internal energies around 40 kJcm^{-3}, the SESAME pressure is significantly higher than the QMD predictions. According to Ref. [173], this is due to an overestimation of the degree of dissociation in the SESAME table. For higher energy densities, where water is largely dissociated, the agreement with the molecular dynamics calculations is again very good. The extrapolated IAPWS-95 formulation agrees well with the QMD simulations for $U \leq 50$ kJcm^{-3}, but the predicted pressure values are too low for $U > 50$ kJcm^{-3} when water is largely dissociated, which is not considered in IAPWS-95. Since experimental U_{avg} data obtained in the present paper stay below 50 kJcm^{-3} and because relatively few data points are available from the QMD simulations, we use the IAPWS-95 formulation for the interpretation of these data.

3.7. Demonstration of micro-material-processing capability of slm UV nanosecond pulses

Scenario 3 of section II.B suggests that the energy deposition by UV ns pulses with smooth temporal shape features a tunable nanoregime and an also tunable but more disruptive micro-regime, with a distinct threshold between them. The existence of both regimes and their potential for material processing in transparent dielectrics is demonstrated by producing dissections in cornea such as needed for flap creation in LASIK refractive surgery [13,17,24,105], and by creating micro-patterns and refractive index changes in borosilicate crown glass (BK7). For this purpose, single-longitudinal mode (slm) laser pulses with 355 nm wavelength and 0.56 ns duration were focused into the samples through a Zeiss LD Plan-Neofluar 63x/0.75corr objective, with the correction ring adjusted to the desired focusing depth. Samples were moved laterally using a computer-controlled translation stage programmed in LabView, and the focal depth was adjusted using a piezoelectric objective positioner (PI P-725.4CD PIFOC).

For the experiments on cornea, porcine eyes were obtained from a slaughterhouse and stored in physiological saline at 4°C until use, no longer than 4h after enucleation. Specimens with 9 mm diameter were extracted using a trephine and placed in a sample holder that was capped with a microscope cover glass. The specimens were immersed in physiological saline for index matching purposes. To mimic conditions in refractive LASIK surgery, the numerical aperture was reduced to 0.38 by inserting a mask into the objective's rear aperture.

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